MISB MOTION IMAGERY STANDARDS BOARD	MISB ST 0801.9
STANDARD	**21 October 2021**
Photogrammetry Metadata for Motion Imagery	

1 Scope

MISB ST 0801 defines metadata necessary to support precise geopositioning for photogrammetric exploitation of Motion Imagery. The ability to geo-locate points on an image with known confidence is an important capability.

Other standards can reference the metadata items in this document by including these items in their respective data sets (e.g., truncation pack, floating length pack, Local Set, etc.). This standard does not address metadata necessary for the primary exploitation of the Motion Imagery (including such elements as mission number, sensor type, platform type, etc.) and security metadata.

Populating photogrammetry metadata at the earliest point in the image workflow maximizes fidelity. In most cases, this is on platform at the Motion Imagery sensor, although post processing may improve point-positioning accuracy.

2 References

MISB references cited herein reflect versions current to the publication date of a document. In the event of a MISB document correction, the corrected document will have a single letter Minor Version appended to the Major Version number per the MISB Document Development Process [1]. For example, corrections to ST 0001.2, which has a Major Version of 2, result in a new version titled ST 0001.2a, which includes a Minor Version of "a". The MISB will not update the referring document (this document) with the Minor Version number change. When acquiring any MISB reference listed below from an NGA repository the latest version may be either a Major or Minor Version.

[1] MISB Document Development Process, Feb 2020.

[2] MISB ST 0107.5 KLV Metadata in Motion Imagery, Oct 2021.

[3] MISB ST 1201.5 Floating Point to Integer Mapping, Jun 2021.

[4] "Coordinate Systems," [Online]. Available: http://earth-info.nga.mil/GandG/update/index.php?dir=coordsys&action=coordsys.

[5] NIMA TR8350.2: Department of Defense World Geodetic System 1984, Its Definitions and Relationships with Local Geodetic Systems, 23 Jun 2004.

[6] MISB ST 1202.2 Generalized Transformation Parameters, Feb 2015.

[7] MISB ST 1010.3 Generalized Standard Deviation and Correlation Coefficient Metadata, Oct 2016.

3 Revision History

Revision	Date	Summary of Changes
ST 0801.9	10/21/2021	• Removed KLV UL's (keys) from Tables • Minor editorial clean up throughout document • Changed to use Azimuth versus Heading consistently • Changed Table 13 to correctly use Line/Sample versus Row/Column • Added Sensor Calibrated / Effective Focal Length Item • Removed IMAP lengths – implementing systems determine IMAP lengths to maintain desired precision • Updated References [2], [3] • Deprecated Reqs -02, -11 • Changed document title

4 Acronyms and Definitions

CSM	Community Sensor Model
ECEF	Earth-Centered, Earth Fixed
GPS	Global Positioning System
IMU	Inertial Measurement Unit
KLV	Key-Length-Value
LRCS	Local Reference Coordinate System
LRP	Local Reference Plane
LRF	Laser Range Finder
MISB	Motion Imagery Standards Board
NED	North-East-Down
NED-LRCS	NED Local Reference Coordinate System
NED-LRP	NED Local Reference Plane
RP	Recommended Practice
SI	International System of Units
SMPTE	Society of Motion Picture and Television Engineers
ST	Standard
WGS-84	World Geodetic System of 1984

5 Introduction

This standard defines Motion Imagery photogrammetric metadata to enable frame-accurate exploitation.

5.1 Frame Sensor Model Overview

A Frame Sensor Model provides the mathematical relationship between an object of interest in the physical world and the image of the object taken with a frame-based camera or sensor. This section defines the terms the mathematical relationships use; Section 6 defines the actual metadata items, units, and data formats.

Figure 1 illustrates an ideal camera (i.e., sensor) imaging a *scene*, which is the space in the physical world "sensed" to form an Image. An ideal camera (and its model) does not have optical distortions or other adjustments. The illustration shows the sensor positioned above the scene looking downward. The Sensor Reference Point is an arbitrary point on or about the sensing device, which defines the position of the physical sensor. The Sensor Reference Point is measurable (e.g., GPS) and unambiguously identified with the physical sensor. Scene energy (e.g., reflected light) comes from the scene and enters the sensor, which records the image. The sensor model uses single perspective geometry when defining the mathematical relationship between the scene and image; therefore, the illustration shows the Perspective Center, which is the point all scene energy passes through before collection by the sensor's Detector Array. The illustration also indicates the four corners of the scene and their standard numbering order (upper-left, upper-right, lower-right, and lower-left).

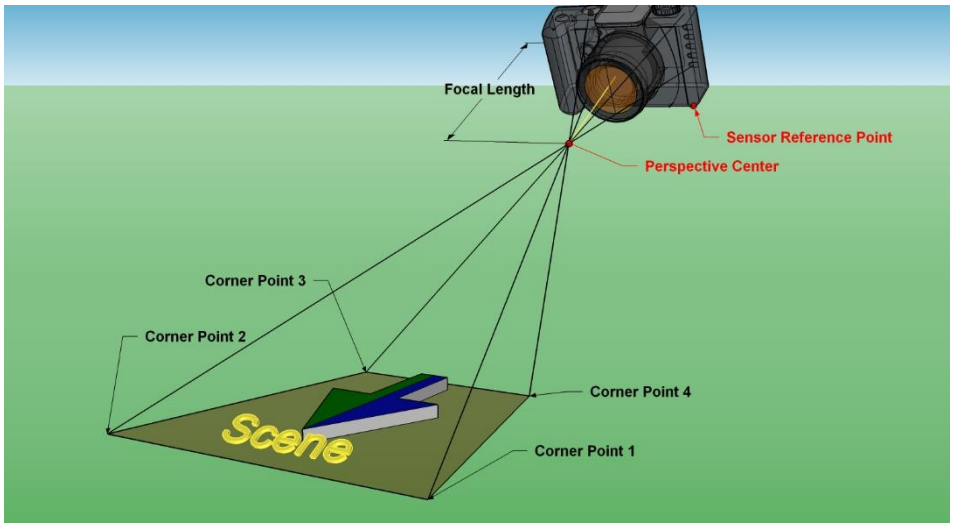


Figure 1: Overview of Sensor and Scene Relationship

Figure 2 illustrates a Detector Array inside the sensor. The Detector Array is a two-dimensional grid of individual detectors each measuring its local portion of the scene energy, the combination of which becomes the image. With an ideal camera (e.g., no distortions), the line orthogonal to the Detector Array which passes through the Perspective Center is the Principal Axis (referred to in this document as the *boresight*). The distance from the Perspective Center to the Detector Array is the Focal Length. Figure 2 shows the Sensor Reference Point is arbitrary and does not necessarily coincide with a specific point in the Detector Array nor the Perspective Center. Additional metadata provides the relationship between the Sensor Reference Point's position and the Perspective Center. The Sensor Reference Point defines the position of the sensor system and is the reference point for its velocity.

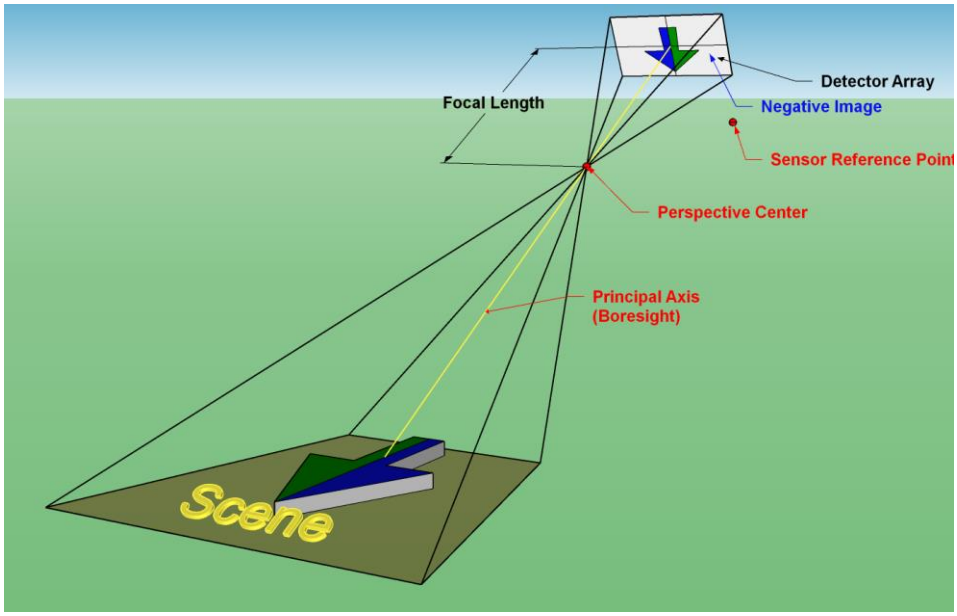


Figure 2: Sensor Detector Array

The Detector Array records a negative image (i.e., flipped top to bottom and left to right) because all scene energy passes through the Perspective Center. Figure 2 illustrates the Negative Image by showing the blue-green arrow inverted and flipped. Many sensor models use the positive image which is located the same distance (the Focal Length) from the Perspective Center but between the Perspective Center and scene. Figure 3 illustrates the Positive Image at an equivalent focal length from the Perspective Center toward the Scene; the blue-green arrow shows the positive image orientation matches the scenes orientation. The mathematics of using the positive image are equivalent to using the reflection of the negative image.

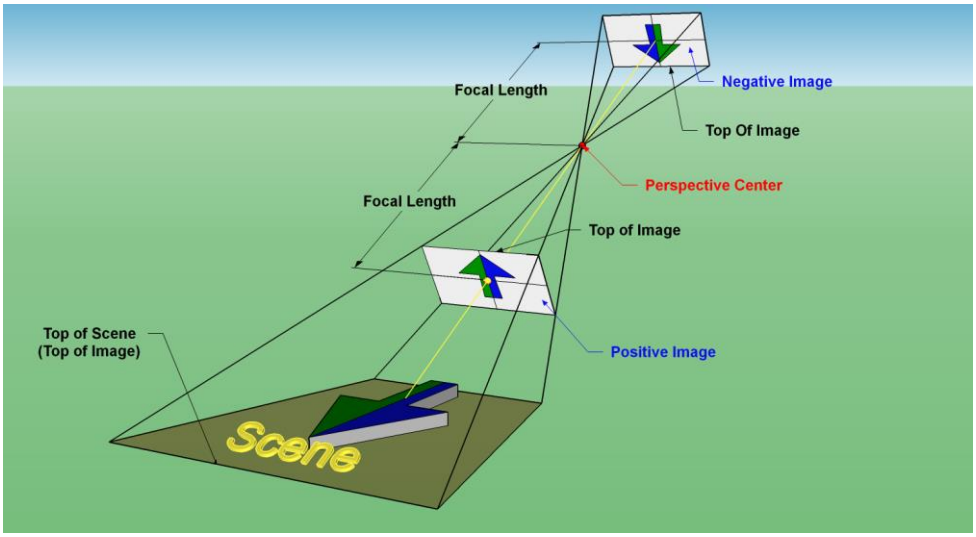


Figure 3: Negative and Positive Images

Figure 4 illustrates the model's points of interest, some of which are the origin of a frame of reference (right-handed orientation). The red, green, and blue (RGB) vectors illustrate the x, y,

and z axes, respectively, of each orientation. Each point uses a different frame of reference for their measurements, so normalizing the positions to the same frame of reference is inherent in the modeling process. The illustration shows the Target Vector, which is the vector from the Perspective Center to the Target Point.

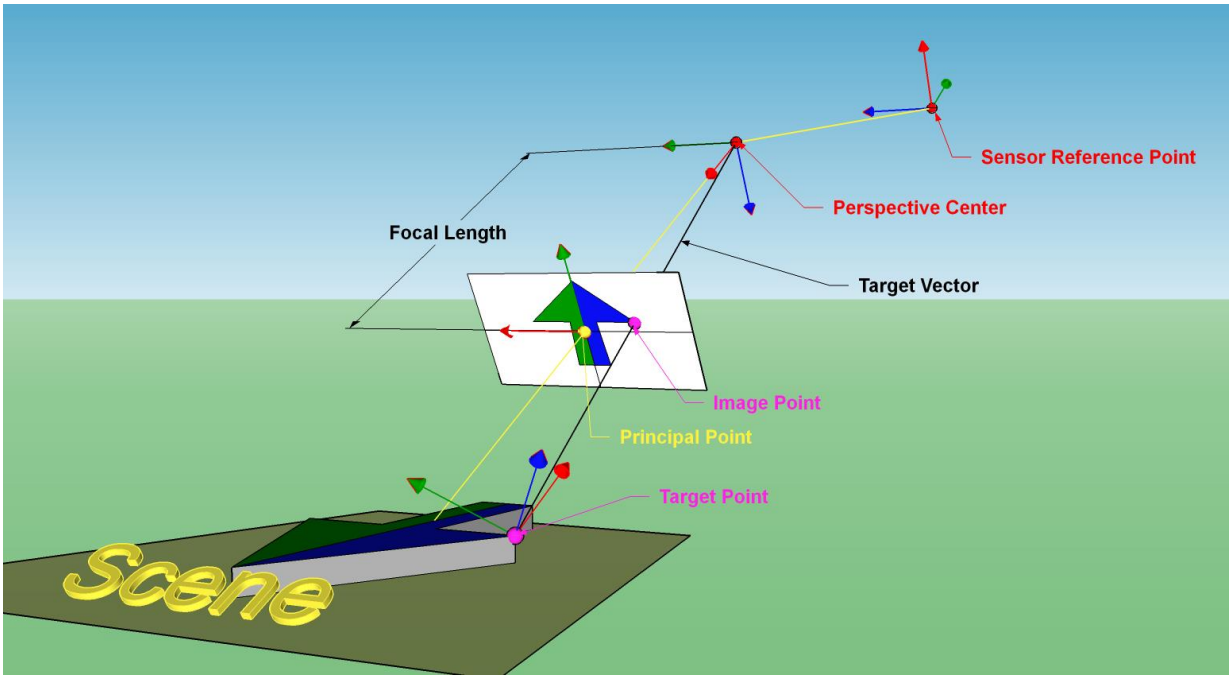


Figure 4: Model Frames of Reference

- The **Sensor Reference Point** is an absolute geodetic location (latitude, longitude, and height above ellipsoid). The Sensor Reference Point is the origin of a frame of reference having its orientation measured relative to North-East-Down (NED). The Sensor Reference Point provides the origin and orientation of the frame of reference for defining the position of the Perspective Center. Section 6.3 provides details for the Sensor Reference Point's position, velocity, orientation, and orientation velocities. Equation 5 uses the term *IMU* (Inertial Measurement Unit) as the name for this frame of reference.
- The **Perspective Center** is a position and orientation measured relative to the Sensor Reference Point's frame of reference. The orientation of the Perspective Center axes has the positive x-axis (red) aligning with the principal axis pointing toward the scene; the positive y-axis (green) pointing in the direction to the right side of the scene (as viewed by the image); and the positive z-axis (blue) pointing toward the bottom of the (positive) image, which completes a right-handed coordinate system. Section 6.5.2 provides details for the Perspective Center's position, and axes orientations. Section 6.5.2 uses the term Boresight Offset to define the location of the Perspective Center relative to the Sensor Position; additionally, the term Boresight Delta Angle describes the orientation of the Perspective Center axes (or x, y, and z -axes) relative to the Sensor Reference Point and its axes. Equation 5 uses the term *LOS* (Line of Sight) as the name for this frame of reference.

- With an ideal camera the **Principal Point** (in yellow) is located at the center of the image and forms the origin of the Principal Point frame of reference. The orientation of the image's frame of reference has the positive x-axis (red) pointing toward the right side of the image, the y-axis (green) pointing towards the top of the (positive) image and the z-axis (not visible in the figure) orthogonal to the image plane, pointing out of the image toward the Perspective Center (completing a right-handed coordinate system). Section 6.5.4 provides details for the Principal Point Offset and Focal Length. Equation 5 uses the term *Frame* as the name for this frame of reference.
- The **Image Point** is the point of intersection between the Target Vector and the focal plane. The Image Point position location is relative to the image's Principal Point.
- The **Target Point** is a position of an object in Earth Centered Earth Fixed (ECEF) coordinates. Figure 4 illustrates the ECEF x, y, and z axes for the given Target Point.

The ideal camera is the starting point for the mathematical relationship between an object of interest and the image. Model adjustments correct for real-world realities due to manufacturing and environmental impacts. One adjustment is the location of the Principal Point. The Principal Point is the intersection of the Principal Axis and the image. With an ideal camera the Principal Point is in the exact center of the image. Manufacturing imperfections and other issues may offset the Principal Point from the image's center. The location of the Principal Point is the **Principal Point Offset**. Figure 5 illustrates the Principal Point shifted from the Center Of Frame; the Principal Axis (yellow line) intersects the image at the Principal Point, shown as the yellow dot on the frame. The Principal Point Offset is x and y distance from the Center of Frame point to the Principal Point.

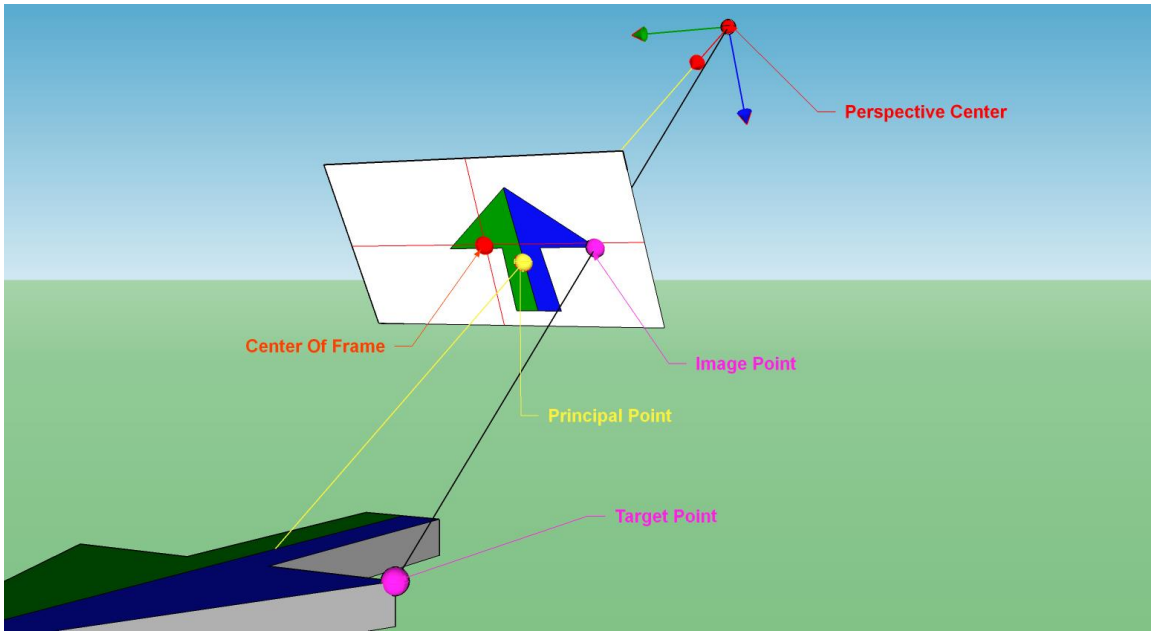


Figure 5: Principal Point Offset

6 Photogrammetry Metadata Groups

This document organizes photogrammetric metadata into photogrammetric metadata groups. Each group follows the conventions in Section 6.1 with an associated table to identify specific metadata items as Section 6.2 describes.

6.1 Conventions

All metadata items within this document follow the MISB's rules on formatting. MISB ST 0107 [2] provides usage rules for metadata.

Requirement(s)	
ST 0801.5-01	All KLV-encoded metadata shall be expressed in accordance with MISB ST 0107.
ST 0801.5-03	Measurements shall be expressed using the International System of Units (SI).

To promote bit efficiency, floating point values may be "packed" as integers, in accordance with MISB ST 1201 [3].

All angle measurements are in half circles. To obtain the value of an angle in radians, multiply the number of half circles in the measurement by pi (π). Angle measurements are packed as integer representations of rational numbers.

This standard defines pi (π) as: π = 3.1415926535 8979324

This is the value of pi used in the coordinate conversion software developed and controlled by NGA; specifically, MSP GEOTRANS 3.0 (Geographic Translator) [4].

Requirement	
ST 0801.5-04	When converting values between radians and half circles, the value of pi (π) shall be 3.1415926535 8979324.

6.2 Metadata Tables

The metadata tables below describe in detail the items of Photogrammetry Metadata for Motion Imagery.

Each table of metadata items contains the following columns:

- *Name* is a descriptive name for the metadata item
- *Units* indicated the units of measure in SI units, if appropriate
- *Format* indicates the Value encoding method, which can be one of the following:
    - Unsigned integer, or uint, notation, see MISB ST 0107
        - Example: uint64 is an unsigned integer of 64-bits
    - IMAP notation, for mapping ("packing") a floating-point value into an integer value; see MISB ST 1201
        - IMAPB(<min float>, <max float>)
        - Example: IMAPB(-1234,+1234) indicates that a floating-point value within the range from -1234.0 to +1234.0 maps to an integer. The length (i.e., number of bytes) of the mapping is determined by a parent standard or at runtime.

- float notation, see ST 0107
  - Example: float32 is a 32-bit IEEE floating point number

6.3 Organization of the Photogrammetry Metadata

This document organizes photogrammetric metadata into three photogrammetric metadata groups: Photogrammetry External Parameters, Photogrammetry Internal Parameters, and Miscellaneous Parameters.

1) Photogrammetry External Parameters (see Section 6.4)
   a. Sensor Position
   b. Sensor Velocity
   c. Sensor Absolute Orientation
   d. Sensor Absolute Orientation Rate
2) Photogrammetry Internal Parameters (see Section 6.5)
   a. Boresight
   b. Image Size
   c. Focal Plane
   d. Radial Distortion
   e. Tangential-Decentering
   f. Affine Correction
3) Miscellaneous Parameters (see Section 6.6)
   a. Slant Range

6.4 Photogrammetry External Parameters

The Photogrammetry External Parameters metadata group relates the sensor to the "real world" using the World Geodetic System-1984 (WGS-84) coordinate frame. All the position coordinates and velocity metadata are with respect to this coordinate reference.

This standard specifies using a WGS-84 coordinate Cartesian, Earth-Centered, Earth-Fixed (ECEF) coordinate system. Figure 6 illustrates the earth's ellipsoid with the ECEF origin (0, 0, 0) and axes in green. The ECEF's origin is the earth's center of mass. The ECEF x-axis points to the intersection of the equator and the IERS Reference Meridian (IRM) (Greenwich meridian), the z-axis points towards the North Pole (true North), and the y-axis completes the right-handed coordinate system [5]. An example ECEF position shows its x, y, and z coordinates. All ECEF measurements use the International System of Units (SI).

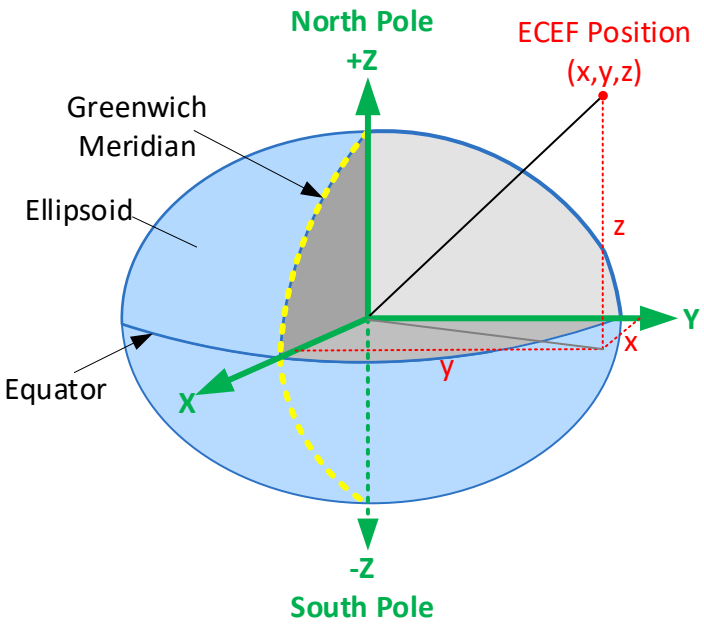


Figure 6: ECEF Coordinate System

Requirement	
ST 0801.5-05	Sensor position and velocity shall be expressed using the WGS-84 Earth-Centered, Earth-Fixed (ECEF) coordinate frame.

This coordinate system is consistent with the native coordinate format of the Global Positioning System (GPS). When transforming these coordinates to other systems (e.g., Latitude, Longitude, and Height-Above-Ellipsoid), use techniques which minimize numerical errors.

This standard specifies the use of sensor orientation relative to a "local" coordinate frame using a geodetic North-East-Down (NED) system at the ECEF location of the sensor. Figure 7 illustrates the NED local coordinate system with the X-Y plane (yellow) parallel to a plane (blue) tangent to the ellipsoid. The X-axis aligns with the longitudinal lines with positive values going north, and the Y-axis aligns with the latitude lines with positive values going east. The Z-axis of each plane is mutually orthogonal to the X and Y axes with positive Z values increasing in the downwards direction geodetically toward the earth's Z axis.

In this standard, the X-Y plane (yellow) is the NED Local Reference Plane (NED-LRP), with the term NED Local Reference Coordinate System (NED-LRCS) denoting all three axes. The Z axes of both X-Y planes are collinear.

Section 7 provides additional details of sensor position and orientation along with corresponding figures and rotation angles relative to the local coordinate system, where they are applied sequentially as azimuth-pitch-roll.

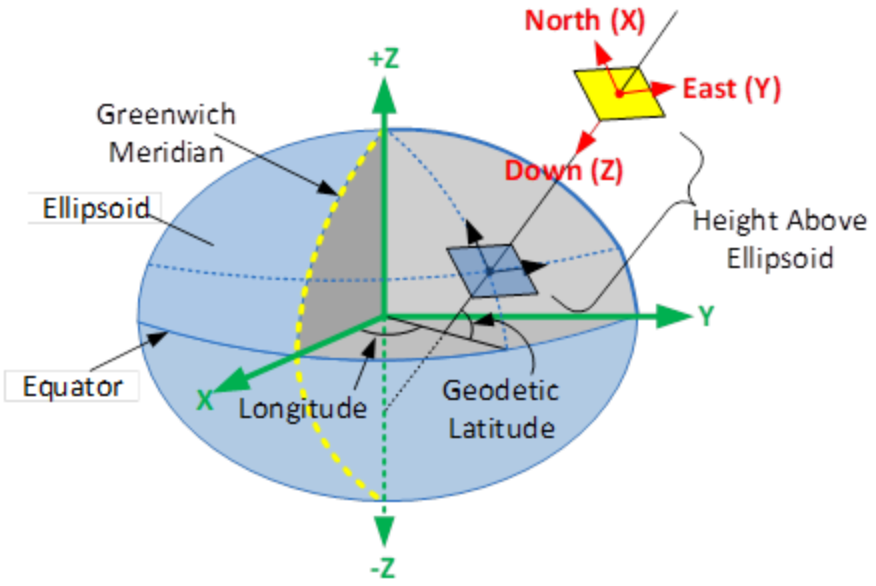


Figure 7: The NED Local Reference Coordinate System (NED-LRCS)

Requirement	
ST 0801.5-06	Sensor orientation shall be expressed using a North-East-Down (NED) coordinate frame located at the Earth-Centered, Earth-Fixed (ECEF) position of the sensor.

6.4.1 Sensor Position Metadata

The Sensor Position metadata group provides the location of the sensor reference point within a sensor. Figure 1, Figure 2, and Figure 4 illustrate one example of a Sensor Reference Point on the corner of a camera. Another example of a reference point is the center of rotation of the sensor (i.e., the point about which a two- or three-axis gimbal rotates). The center of rotation is a point of convenience because its location does not change depending on the sensor orientation relative to the platform's reference frame. Table 1 lists the Sensor Position metadata which is the ECEF coordinates of the sensor reference point.

Table 1: Sensor Position Metadata

Name	Units	Format	Math Symbol
Sensor ECEF Position Component X	m	IMAPB(-1e9, 1e9)	X_{Sen}
Sensor ECEF Position Component Y	m	IMAPB(-1e9, 1e9)	Y_{Sen}
Sensor ECEF Position Component Z	m	IMAPB(-1e9, 1e9)	Z_{Sen}

The intended use of this metadata group is to perform photogrammetric computations, which are based on the sensor perspective center. (The sensor perspective center is analogous to the pinhole of a pinhole camera.) This metadata group alone does not describe the sensor perspective center, but when used with the Boresight metadata group, described in Section 6.5.2, it gives the exact location needed in the photogrammetric computations.

6.4.2 Sensor Velocity Metadata

Table 2 lists the Sensor Velocity metadata which is the ECEF velocity components of the sensor reference point.

Table 2: Sensor Velocity Metadata

Name	Units	Format
Sensor ECEF Velocity Component X	m/s	IMAPB(-25e3, 25e3)
Sensor ECEF Velocity Component Y	m/s	IMAPB(-25e3, 25e3)
Sensor ECEF Velocity Component Z	m/s	IMAPB(-25e3, 25e3)

6.4.3 Sensor Absolute Orientation Metadata

The Sensor Absolute Orientation metadata group provides the sensor Azimuth, Pitch, and Roll angles (azimuth is also known as "heading".) These items specify sensor orientation with respect to a North-East-Down frame of reference located at the sensor perspective center, see Figure 4. These items describe the direction in which the sensor "Reference Axis" is pointing.

The combination of the sensor Reference Axis and the Boresight metadata group, described in Section 6.5.2, defines the sensor Principal Axis, also known as the sensor line-of-sight axis, or boresight. A detailed explanation appears in Section 7.

The Azimuth of a sensor is the angle from True North to the boresight vector projected onto the local horizontal plane. Range of values is 0 to (almost) 2 half-circles; North is 0, East is 0.5 half-circles; South is 1 half-circle, and West is 1.5 half-circles.

The Pitch of a sensor describes the angle its boresight vector makes with the horizontal, where the vertical is perpendicular to the ellipsoid; positive (negative) angles describe a nose up (down) orientation. Range of values is -1.0 half-circles to +1.0 half-circles.

The Roll of a sensor is the rotation angle about the boresight vector which completes the sensor orientation. A clockwise roll angle is a positive measure. The range of values is -1.0 to +1.0 half-circles.

System developers use the azimuth-pitch-roll angles to build rotation matrices, as described in Section 7.

The azimuth-pitch-roll formulation of the rotation matrix has a known singularity point where the pitch angle is equal to +90 degrees or -90 degrees. During calculation, use caution to avoid errors as the sensor passes through these discontinuity points, see Section 7 for more details.

Note that azimuth, pitch, and roll must be applied in a strict sequence, as described in Figure 12, Figure 13, and Figure 14 of Section 7.

Requirement	
ST 0801.5-07	During coordinate system transformation calculations, sensor rotation angles shall be applied in the sequence (1) azimuth, (2) pitch, and (3) roll.

Table 3 lists the sensor absolute orientation metadata.

Table 3: Sensor Absolute Orientation Metadata

Name	Units	Format	Math Symbol
Sensor Absolute Azimuth	half circles	IMAPB(0, 2)	Az
Sensor Absolute Pitch	half circles	IMAPB(-1, 1)	Pt
Sensor Absolute Roll	half circles	IMAPB(-1, 1)	Ro

6.4.4 Sensor Absolute Orientation Rate Metadata

The Sensor Absolute Orientation Rate metadata group provides the velocities of the rotations around the sensor's axes (in half-circles per second). Table 4 lists the sensor absolute orientation rate metadata. The definitions and sign conventions for this metadata group (time rate of change) are the same as those given in Section 6.4.3.

Table 4: Sensor Absolute Orientation Rate Metadata

Name	Units	Format
Sensor Absolute Azimuth Rate	half circles /sec	IMAPB(-1, 1)
Sensor Absolute Pitch Rate	half circles /sec	IMAPB(-1, 1)
Sensor Absolute Roll Rate	half circles /sec	IMAPB(-1, 1)

6.5 Photogrammetry Internal Parameters

Within photogrammetry, data values representing the configuration and orientation of optical sensor/detector systems behind a lens or aperture are the internal parameters. The Photogrammetry Internal Parameters represent a full and complete description of the internal sensor geometry.

Motion Imagery photogrammetry applications use the internal parameters to perform transformations between image coordinate systems. The metadata definitions for the internal parameters described here provide representation of known systematic errors in the imaging system. The transformation between image coordinate systems must use coordinates referenced to the original full image resolution (i.e., no digital zooming).

Requirement	
ST 0801.5-08	Transformation between image coordinate systems shall use the original full image resolution (i.e., no digital zooming).

6.5.1 Image Coordinate Systems

This standard uses three image-based coordinate systems for physical focal plane arrays and digital images: Row and Column; Line and Sample; and Measured (x, y). The first two use units of pixels, and the third uses physical measures, such as, millimeters. The Community Sensor Model (CSM) standard coordinate system is the Line and Sample coordinate system. Table 5 summarizes the characteristics of each coordinate system.

Table 5: Image Coordinate Systems

Coordinate System	Units	Format	Origin
Row and Column	Whole Pixels	Integer Value	Upper Left Pixel
Line and Sample	Pixels	Real Value	Upper Left corner of Pixel Array
Measured	mm	Real Value	Principal Point

6.5.1.1 Row and Column Coordinate System

The **Row and Column Coordinate** (R, C) system is an integer-based coordinate system which indexes individual pixels in the image. The Row and Column Coordinate System's origin $(0,0)_{R,C}$ is the pixel at the upper left corner of the image with the positive Row axis pointing downwards and the positive Column axis pointing towards the right. Figure 8 illustrates the Row and Column Coordinate System overlaid on a five row by seven column (5 x 7) image. The Row and Column axes and the origin are blue with the circles denoting the discrete integer coordinates. A sample coordinate at row 2, column 3 shows the coordinates are in the center of the pixel.

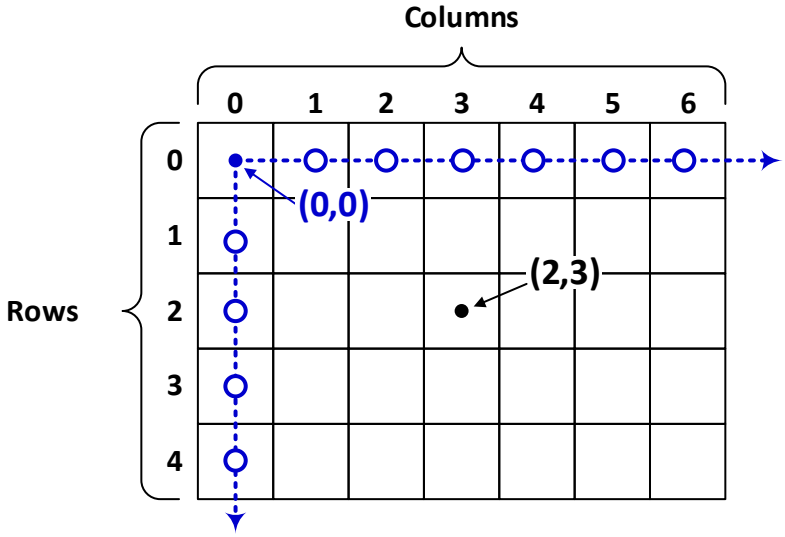


Figure 8: Row and Column Coordinates

6.5.1.2 Line and Sample Coordinate System

The **Line and Sample Coordinate** (L, S) system (CSM's Image Coordinate System) is a real-valued coordinate system which measures a position in the image in decimal pixels. The Line and Sample Coordinate System's origin $(0.0, 0.0)_{L,S}$ is at the upper left corner of the upper left pixel of the image with the positive Line axis pointing downwards and the positive Sample axis pointing towards the right. The center of each pixel is at half pixel high and wide in both the Line and Sample directions. For example, the center of the upper left pixel has line, sample coordinates of (0.5, 0.5). Figure 9 illustrates the Line and Sample Coordinate System overlaid on a five row by seven column (5 x 7) image; the Line and Sample axes and the origin are red. A

sample coordinate in the same position as Figure 8 is at line 2.5, sample 3.5 which is in the center of the pixel.

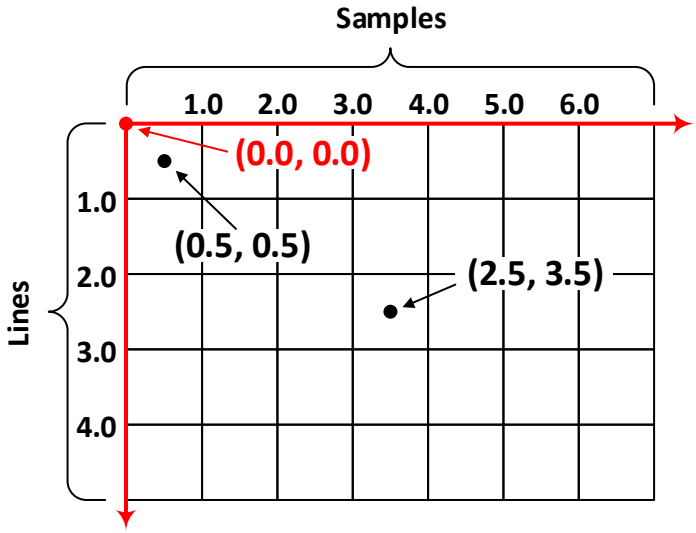


Figure 9: Line and Sample Coordinates

Figure 10 is an image with a Line and Sample axis overlay.

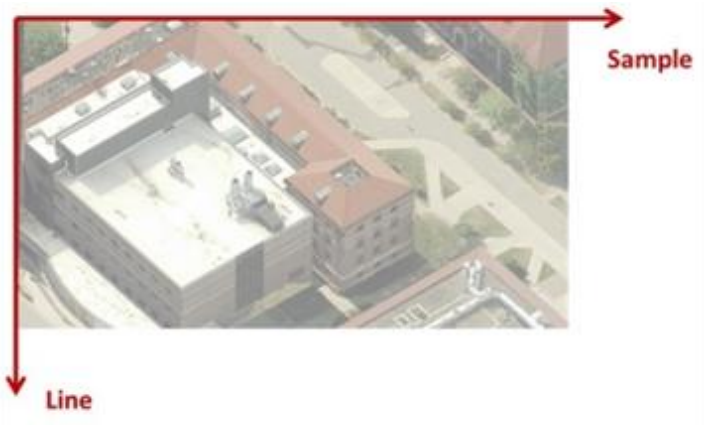


Figure 10: Line and Sample Axes Overlay on Real Image

6.5.1.3 Measured Coordinate System

The **Measured Coordinate System** (x, y) is a real-valued coordinate system which measures an image position from the center of the image in millimeters; the center of the image is the origin of the Measured Coordinate System. The origin is located at the line/sample position when dividing the number of rows ($N_R$in Table 8) by 2 and dividing the number of columns (N_C in Table 8) by 2. Equation 1 shows the computation of the Measured Coordinate System's origin, $(0.0, 0.0)_{x,y}$, in line/sample coordinates, note the reversal to sample/line (S, L) to match the proper directions for x and y.

$$(0.0, 0.0)_{x,y} = (S, L) = \left(\frac{N_C}{2}, \frac{N_R}{2}\right) \qquad \textbf{Equation 1}$$

The x, y coordinates are measures from the origin using multiples of the pixel size x (p_x in Table 8) and pixel size y (p_y in Table 8). Figure 11 illustrates the Measured Coordinate system with the origin and axes in green and using a pixel size of 5.1 by 6.2 μm, or 0.00051 by 0.00062 millimeters. A sample coordinate in the same position as Figure 9 is at (x, y) = (-15.3, 12.4) which is in the center of the pixel.

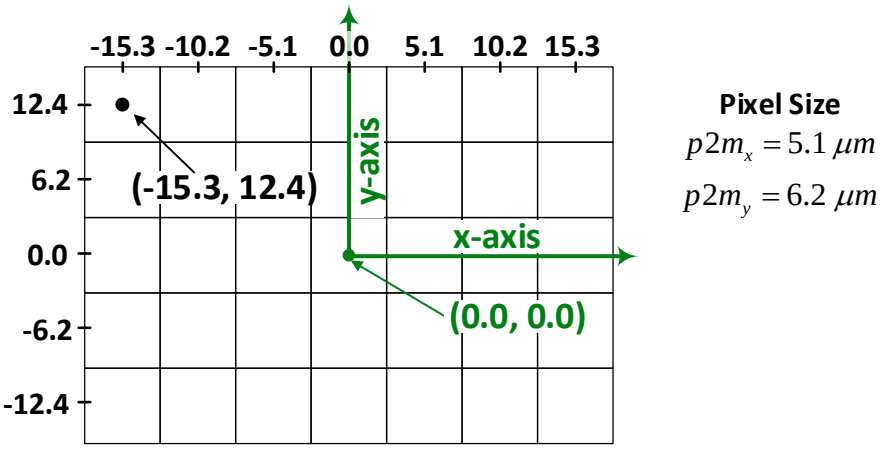


Figure 11: Measured Coordinate System

6.5.1.4 Coordinate Transformations

Table 6 lists the coordinate transform between each of the image coordinate systems. The table uses the floor operation (i.e., $\lfloor value \rfloor$) when converting to Row and Column.

Table 6:Transformation Equations

To / From	Row and Column (R, C) = …	Line and Sample (L, S) = …	Measured (x, y) = …
Row (R) and Column (C)		$R + \frac{1}{2}, C + \frac{1}{2}$	$\left(C + \frac{1}{2} - \frac{N_C}{2}\right)p_x, \left(\frac{N_R}{2} - R - \frac{1}{2}\right)p_y$
Line (L) and Sample (S)	$\lfloor L \rfloor, \lfloor S \rfloor$		$\left(S - \frac{N_C}{2}\right)p_x, \left(\frac{N_R}{2} - L\right)p_y$
Measured (x, y)	$\left\lfloor \frac{N_R}{2} - \frac{y}{p_y} \right\rfloor, \left\lfloor \frac{x}{p_x} + \frac{N_C}{2} \right\rfloor$	$\frac{N_R}{2} - \frac{y}{p_y}, \frac{x}{p_x} + \frac{N_C}{2}$	

MISB ST 1202 [6] defines additional forms of pixel to image mappings for chips, digitally zoomed images, or other types of general transformations.

6.5.2 Boresight Metadata

The Boresight Offset Delta X, Delta Y, and Delta Z metadata group provides items for the mathematical translation from the origin of the Sensor Reference Point to the origin of the sensor perspective center.

The rotational offsets Delta Angle 1, Delta Angle 2, and Delta Angle 3 are angular rotations applied to the sensor Reference Axes to align with the sensor Principal Axes. Delta Angle 1 represents the rotation about the twice-rotated x-axis of the sensor reference frame. Delta Angle 2 represents the rotation about the once-rotated y-axis of the sensor reference frame. Delta Angle 3 represents the rotation about the z-axis of the sensor reference frame.

Apply the rotations in the following order: (1) apply the Delta Angle 3 rotation about the z-axis of the sensor reference frame; (2) next apply the Delta Angle 2 rotation about the once-rotated y-axis of the sensor reference frame; and (3) finally apply the Delta Angle 1 rotation about the twice-rotated x-axis of the sensor reference frame. These sequential rotations rotate the measured sensor reference frame to the sensor optical line of sight reference frame. If the sensor reference frame aligns with the sensor optical axis, the three boresight angles equal to zero.

See Section 7 for figures describing the rotations and for further information on the application of the boresight offsets. Table 7 lists the boresight metadata.

Table 7: Boresight Metadata

Name	Units	Format	Math Symbol
Boresight Offset Delta X	m	IMAPB(-300, 300)	$b_{\Delta X}$
Boresight Offset Delta Y	m	IMAPB(-300, 300)	$b_{\Delta Y}$
Boresight Offset Delta Z	m	IMAPB(-300, 300)	$b_{\Delta Z}$
Boresight Delta Angle 1	half circles	IMAPB(-0.25, 0.25)	β_1
Boresight Delta Angle 2	half circles	IMAPB(-0.25, 0.25)	β_2
Boresight Delta Angle 3	half circles	IMAPB(-0.25, 0.25)	β_3

6.5.3 Image Size Metadata

The Image Size metadata group provides the number of image rows (i.e., the image height), the number of image columns (i.e., the image width), the size of the pixels in the x-direction, and the size of the pixels in the y-direction. Table 6 provides the transformations between image coordinate systems. Table 8 lists the image size metadata.

Table 8: Image Size Metadata

Name	Units	Format	Math Symbol
Image Rows	pixels	uint16	N_R
Image Columns	pixels	uint16	N_C
Pixel Size X	mm	IMAPB(1e-4, 0.1)	p_x
Pixel Size Y	mm	IMAPB(1e-4, 0.1)	p_y

Requirement	
ST 0801.5-09	When the pixel size in the y direction is unspecified, it shall be assumed equal to the pixel size in the x direction indicating square pixels.

6.5.4 Focal Plane Metadata

The Focal Plane metadata group provides information about the sensor focal plane and imaging geometry. This metadata group contains the Principal Point offset and the effective focal length of the sensor. The principal point offset item $+x_0$ is the Focal Plane Principal Point Offset X and the principal point offset item $+y_0$ is the Focal Plane Principal Point Offset Y. Both x_0 and y_0 items represent principal point offset item in Section 6.5.5 Equation 2 and Section 6.5.7 Equation 4. Table 9 lists the focal plane metadata.

Table 9: Focal Plane Metadata

Name	Units	Format	Math Symbol
Focal Plane Principal Point Offset X	mm	IMAPB(-25, 25)	x_0
Focal Plane Principal Point Offset Y	mm	IMAPB(-25, 25)	y_0
Sensor Calibrated / Effective Focal Length	mm	IMAPB(0, 10 000)	f
Sensor Calibrated / Effective Focal Length Extended	mm	IMAPB(0, 100 000)	f

6.5.5 Radial Distortion Metadata

The Radial Distortion metadata group provides items needed to correct for barrel or pincushion distortions in the sensor optics. Equation 2 models the radial lens distortion d_r for image coordinates x and y as a polynomial function of the radial distance r from the Principal Point. The radial distortion items are the k_0, k_1, k_2, and k_3 constants of Equation 2.

$$d_r = k_0 r + k_1 r^3 + k_2 r^5 + k_3 r^7 \qquad \textbf{Equation 2}$$

where $\bar{x} = x - x_0$ $\bar{y} = y - y_0$ $r = \sqrt{\bar{x}^2 + \bar{y}^2}$

and x_0 and y_0 are the coordinates of the Principal Point.

This model of radial distortion has a limited range for which the residuals of the fit are acceptable. This “valid range” is a Measured Coordinates System distance in mm radially from the principal point. Table 10 lists the radial distortion rate metadata.

Table 10: Radial Distortion Metadata

Name	Units	Format	Math Symbol
Valid Range of Radial Distortion	mm	float32	
Radial Distortion Constant	mm/(mm)	float32	k_0
First Radial Distortion	mm/(mm)^3	float32	k_1
Second Radial Distortion	mm/(mm)^5	float32	k_2
Third Radial Distortion	mm/(mm)^7	float32	k_3

6.5.6 Tangential-Decentering Metadata

The Tangential-Decentering metadata group provides tangential decentering items P_1, P_2, and P_3 to correct image coordinates *x* and *y* for these distortions using Equation 3.

$$\Delta x_{decen} = (1 + P_3 r^2)[P_1(r^2 + 2\bar{x}^2) + 2P_2\bar{x}\,\bar{y}]$$
$$\Delta y_{decen} = (1 + P_3 r^2)[2P_1\bar{x}\bar{y} + P_2(r^2 + 2\bar{y}^2)] \qquad \textbf{Equation 3}$$

where Δx_{decen} and Δy_{decen} are the x and y components of the decentering effect.

$$\bar{x} = x - x_0 \qquad \bar{y} = y - y_0 \qquad r = \sqrt{\bar{x}^2 + \bar{y}^2}$$

where x_0 and y_0 are the coordinates of the Principal Point.

Table 11 lists the tangential-decentering metadata.

Table 11: Tangential-Decentering Metadata

Name	Units	Format	Math Symbol
First Tangential / Decentering	mm/(mm)^2	float32	P_1
Second Tangential / Decentering	mm/(mm)^2	float32	P_2
Third Tangential / Decentering	1/(mm)^2	float32	P_3

6.5.7 Affine Correction Metadata

The Affine Correction metadata group provides affine correction items. Item b_1 is a differential scale correction, and b_2 is a skew correction.

$$\Delta x_{skew-scale} = b_1\bar{x} + b_2\bar{y} \qquad \textbf{Equation 4}$$

where $\bar{x} = x - x_0$ and $\bar{y} = y - y_0$

x_0 and y_0 are the principal point offsets in the x and y directions, respectively; and x and y are the image coordinates in the frame coordinate system. Table 12 lists the affine correction metadata.

Table 12: Affine Correction Metadata

Name	Units	Format	Math Symbol
Differential Scale Affine	mm/mm	float32	b_1
Skewness Affine	mm/mm	float32	b_2

Requirement	
ST 0801.5-10	Where pixels are not square, the differential scale correction, b_1, shall be exactly zero.

6.6 Photogrammetry Miscellaneous Parameters

The following subsection describes parameters that are neither interior nor exterior orientation parameters; however, they are useful when describing the data collected from a Motion Imagery sensor.

6.6.1 Slant Range Metadata

Slant Range is the distance from the perspective center to a point on the ground contained in the scene.

A pedigree component exists to describe the derivation of the slant range value. There are three options: (0) other, indicated by a value of zero; (1) measured, indicated by a value of 1; and (2) computed, indicated by a value of 2.

Typically, a measured range value uses a laser range finder.

Requirement	
ST 0801.5-12	When the slant range pedigree value is absent from the metadata group, but a slant range value is specified, the pedigree value shall be assumed to be 1 (a measured range).

Included in the Slant Range metadata group is a line and sample coordinate for the slant range to indicate the coordinate within the scene for which the value applies. Typically, is at the center of the image.

For measurements obtained using a laser range finder, the metadata group includes a laser range finder (LRF) divergence value to quantify the divergence of the laser range finder. Table 13 lists the slant range metadata.

Table 13: Slant Range Metadata

Name	Units	Format
Slant Range	m	float32
Slant Range Pedigree	N/A	uint(1)
Line Coordinate	pixels	float32
Sample Coordinate	pixels	float32
LRF Divergence	rad	float32

6.7 Error Propagation

This document provides a "library" of metadata items for incorporating into aggregate KLV data structures (e.g., Local Sets) of other specifications (MISB Standards and Recommended Practices). To comply with the objective that these metadata items support precise geolocation, any containing specification must also include appropriate uncertainty information properly formatted as found in MISB ST 1010 [7].

Requirement(s)	
ST 0801.5-13	Specifications that include any of the photogrammetry parameters defined in this specification shall include related variance-covariance uncertainty information.
ST 0801.5-14	Uncertainty information regarding photogrammetry parameters defined in this specification shall be encoded in accordance with MISB ST 1010.

7 Appendix – Rotation Angle and Coordinate Systems Definitions

This appendix provides additional information for applying the information contained in this document.

As a means of providing an overview of the rotation angles and coordinate systems documented in this appendix and their relationships to one another, Equation 5 provides a mathematical description of the projection of ground coordinates (e.g., a target) in the ECEF system into the *Frame's* Measured Coordinate System coordinates.

$$\begin{bmatrix}\tilde{x}\\ \tilde{y}\\ \tilde{z}\end{bmatrix} = \eta R_{LOS}^{Frame} R_{IMU}^{LOS} R_{NED}^{IMU} R_{ECEF}^{NED} \left(\begin{bmatrix}X_T\\ Y_T\\ Z_T\end{bmatrix} - \left(\begin{bmatrix}X_{SEN}\\ Y_{SEN}\\ Z_{SEN}\end{bmatrix} + (R_{NED}^{IMU} R_{ECEF}^{NED})^T \begin{bmatrix}b_{\Delta X}\\ b_{\Delta Y}\\ b_{\Delta Z}\end{bmatrix} \right) \right) \quad \textbf{Equation 5}$$

where $\tilde{x}, \tilde{y}$ are the idealized *Frame* image coordinates (in units of mm) that have undergone the photogrammetric internal corrections documented in Section 6.5,

$\tilde{z} = -f$, where f is the Sensor Calibrated / Effective Focal Length,

η is a scale factor variable cancelled out by dividing the first and second rows by the third row of Equation 5 when exercising the collinearity equations defined by Equation 6, Equation 13 and Equation 8

$$R_{LOS}^{Frame} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{bmatrix} \quad \textbf{Equation 6}$$

X_T, Y_T, Z_T are the ECEF coordinates (in units of meters) of the target,

X_{SEN}, Y_{SEN}, Z_{SEN} are the "Sensor ECEF Position" coordinates (in units of meters) of the center of navigation of the sensor,

$b_{\Delta X}$, $b_{\Delta Y}$, $b_{\Delta Z}$ are the "Boresight Offset Delta" coordinates (in units of meters) in the IMU system,

$$R_{ECEF}^{NED} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\lambda & \sin\lambda & 0 \\ -\sin\lambda & \cos\lambda & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \textbf{Equation 7}$$

where λ and ϕ are the Geodetic latitude and longitude, respectively, of the center of navigation of the sensor (i.e., corresponding to the ECEF coordinates X_{SEN}, Y_{SEN}, Z_{SEN}).

Establishing a convention for rotating a coordinate system to be parallel to another coordinate system requires choosing from a variety of methods that yield the same result. This ST uses the azimuth-pitch-roll sequence for the rotations. To avoid errors in interpretation, define the azimuth, pitch, and roll angles unambiguously with respect to the starting and ending coordinate systems.

This section of the appendix describes the starting coordinate system, the ending coordinate system, and the prescription of the angles used to align the two systems.

The azimuth, pitch, and roll angles are measurements relative to a North-East-Down (NED) system, centered at the sensor perspective center. The NED reference frame is a right-handed coordinate system with North being analogous to the x-axis, East being analogous to the y-axis, and Down being analogous to the z-axis. The destination coordinate system is the Inertial Measurement Unit (IMU) coordinate system, which is a right-handed system with its origin at the center of navigation. For a system with zero boresight offset angles (assume for the remainder of this derivation), the line-of-sight axes align where the x-axis is pointing along the sensor line-of-sight vector, the y-axis is parallel to the rows in the image, and the z-axis is parallel to the columns in the image.

The first angle of rotation aligns the x-axis (North) with the projection of the sensor boresight vector into a horizontal plane by rotating about the NED z-axis (Down). Figure 12 illustrates a rotation around the z-axis, where the x-axis is colored red, the y-axis is colored green, and the z-axis is colored blue. The magnitude of this angle is equal to the azimuth, where a positive angle is in the clockwise direction when looking in the "down" direction; in other words, positive moves the red-axis (x-axis) to the green-axis (y-axis). The angle labeled in the figure, A3, is equivalent to the azimuth.

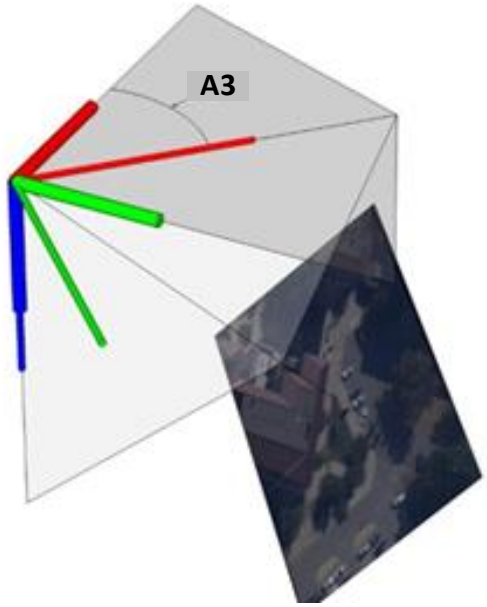


Figure 12: Description of the azimuth rotation

The second rotation points the once-rotated x-axis along the sensor boresight vector by a rotation about the once-rotated y-axis. The magnitude of this angle is the pitch, or a deflection from the local horizon shown in Figure 13. This angle is positive in the up-direction, where the blue-axis (once-rotated z-axis) moves towards the red-axis (once-rotated x-axis). Angle *A2* is equivalent to the sensor pitch and has a negative value when the sensor points towards the Earth.

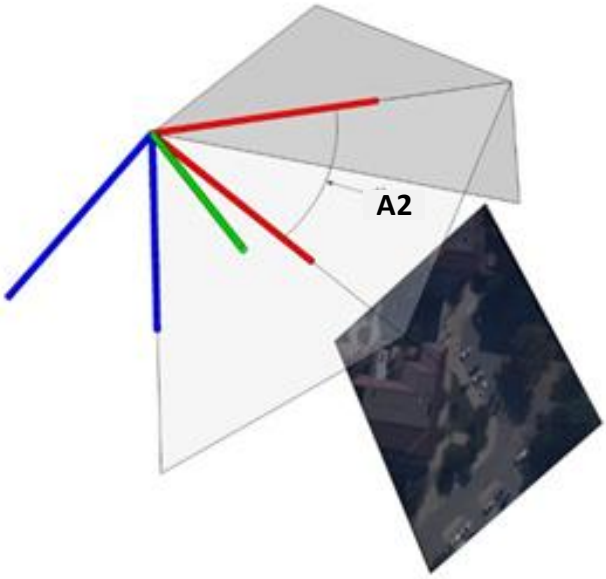


Figure 13: Description of the pitch rotation

The final rotation rotates the image about the sensor boresight vector by a rotation about the twice-rotated x-axis shown in Figure 14. The magnitude of this angle is the roll of the sensor, where it is positive clockwise when looking from the sensor along the boresight vector; in other

words, the green-axis (y-axis) moves towards the blue-axis (twice-rotated z-axis). Angle *A1* is equivalent to the roll angle.

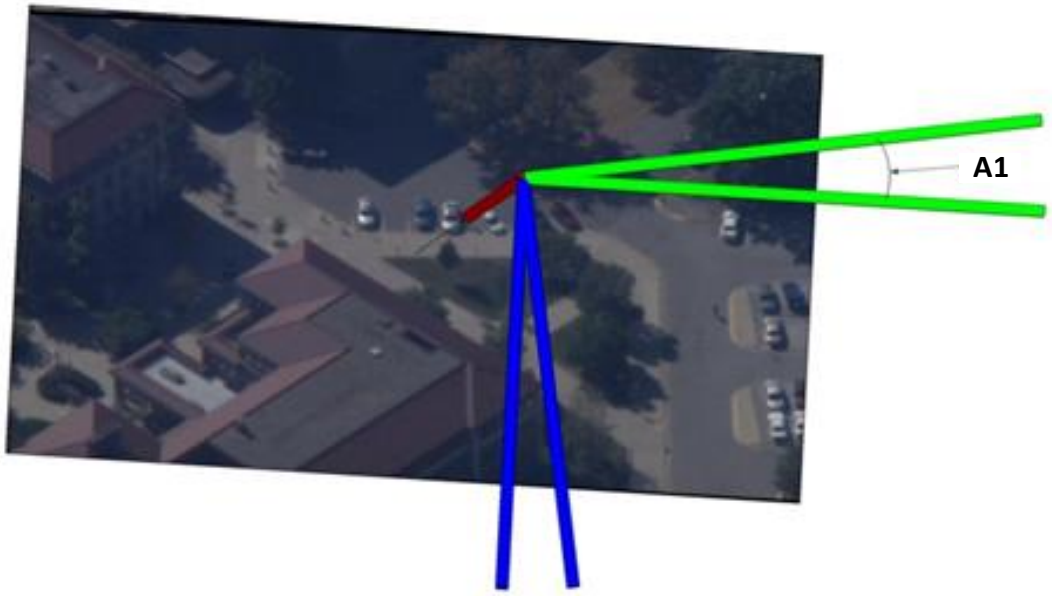


Figure 14: Description of the roll rotation

These rotations align the NED axes parallel to the IMU Axes. Equation 8 describes the total rotation matrix from the NED to the Inertial Measurement Unit (IMU) coordinate system. The IMU coordinate system represents the coordinate system for the senor's Reference Axes.

$$R_{NED}^{IMU} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos Ro & \sin Ro \\ 0 & -\sin Ro & \cos Ro \end{bmatrix} \begin{bmatrix} \cos Pt & 0 & -\sin Pt \\ 0 & 1 & 0 \\ \sin Pt & 0 & \cos Pt \end{bmatrix} \begin{bmatrix} \cos Az & \sin Az & 0 \\ -\sin Az & \cos Az & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \textbf{Equation 8}$$

The values used in this equation for the Azimuth (Az), Pitch (Pt), and Roll (Ro) angles can be derived from any rotation matrix that aligns a NED coordinate system with an IMU coordinate system. These values are consistent regardless of how the initial rotations are defined provided the following prescription is followed. The rotation matrix is refined in Equation 9 through Equation 12 with each of its nine elements labeled.

$$R_{NED}^{IMU} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \quad \textbf{Equation 9}$$

$$A_z = \tan^{-1}\left(\frac{r_{12}}{r_{11}}\right) \quad \textbf{Equation 10}$$

$$P_t = \sin^{-1}(-r_{13}) \quad \textbf{Equation 11}$$

$$R_o = \tan^{-1}\left(\frac{r_{23}}{r_{33}}\right) \quad \textbf{Equation 12}$$

Using this formulation to define the angles allows the data provider to use *any* method of computing the rotation matrix that rotates the NED to the IMU coordinate system without loss of generality. In other words, this formulation does not have underlying assumptions that cause a loss in computational accuracy.

An additional computational note is in dealing with arctangent functions. Since the data collected for the azimuth and roll angles can lie in any of the four quadrants of the unit circle, use the two-argument form of the arctangent function (ATAN2). With the goal of this type of decomposition to obtain an identical rotation matrix, the results of the previously described algorithm satisfy this objective. However, the actual values for the azimuth-pitch-roll may be different. Differences typically occur when the pitch angle is less than -90 degrees or greater than +90 degrees. This condition causes the azimuth to read 180 degrees different from the original angle, and the roll angle also reads 180 degrees different to account for the direction change. Again, the ATAN2 function of the decomposition returns identical results for the reconstructed rotation matrix. There is a possible discontinuity if the goal is to reconstruct the actual angles. To determine the exact angles requires additional information (e.g., adjacent frames or trajectory information).

Similarly, the rotation matrix for the boresight offset angles which rotates the IMU coordinate system to the line-of-sight (Principal) coordinate system uses an identical sequence of rotations as the NED system to the IMU system. Equation 13 describes this rotation matrix.

$$R_{IMU}^{LOS} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\beta_1 & \sin\beta_1 \\ 0 & -\sin\beta_1 & \cos\beta_1 \end{bmatrix} \begin{bmatrix} \cos\beta_2 & 0 & -\sin\beta_2 \\ 0 & 1 & 0 \\ \sin\beta_2 & 0 & \cos\beta_2 \end{bmatrix} \begin{bmatrix} \cos\beta_3 & \sin\beta_3 & 0 \\ -\sin\beta_3 & \cos\beta_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \textbf{Equation 13}$$

Apply the order of rotations similarly to the rotations for the NED to IMU rotations. The previously described Figure 12 through Figure 14 similarly describe the rotation of the IMU to line-of-sight coordinate systems.

After defining all the rotations and coordinate systems used to transform from ground to the image, the following defines how to model the errors in the angle values.

The uncertainty values associated with the “Sensor Absolute Azimuth”, “Sensor Absolute Pitch”, “Sensor Absolute Roll” and their associated correlation coefficients are all with respect to “small angle” errors δAz, δPt and δRo about the sensor coordinate system. These values represent uncertainties in rotation about Z, Y and X axes, respectively, of the sensor coordinate system shown graphically in Figure 15 and mathematically in Equation 14 as the error form of the rotation matrix that aligns the NED system to the IMU system, $R_{NED\,error}^{IMU}$. As the figure and equation show, the sigmas are uncertainties in the Euler angles which define the orientation of the sensor coordinate system with respect to the North-East-Down coordinate system.

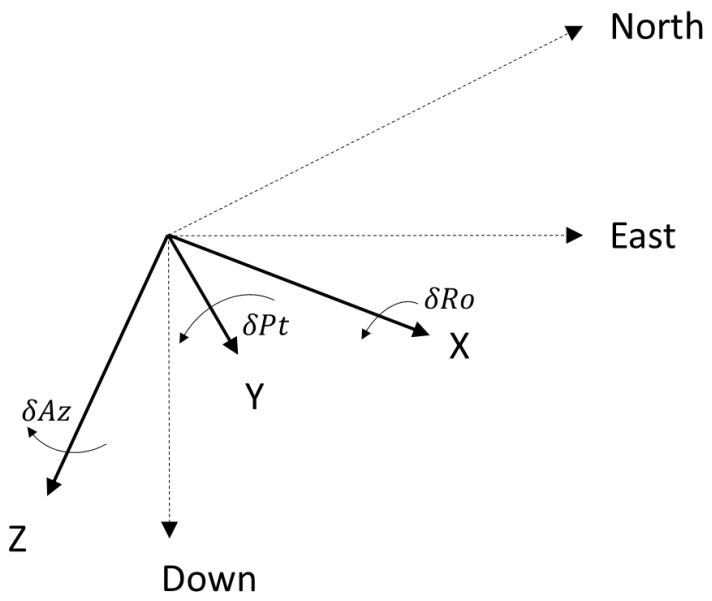


Figure 15: Small Angle errors about the IMU's X, Y and Z axes (North, East, Down axes are shown in dotted lines for context)

$$R^{IMU}_{NED_{error}} = \begin{bmatrix} 1 & \delta Az & -\delta Pt \\ -\delta Az & 1 & \delta Ro \\ \delta Pt & -\delta Ro & 1 \end{bmatrix} R^{IMU}_{NED} \qquad \textbf{Equation 14}$$

The convention for providing sigma and correlation coefficient values for the boresight offset angles is the same as described for sensor absolute angles. The error form of the rotation matrix that aligns the IMU to the LOS system, $R^{LOS}_{IMU_{error}}$, is summarized in Equation 15, where the "small angle" rotations are $\delta\beta_1$, $\delta\beta_2$ and $\delta\beta_3$.

$$R^{LOS}_{IMU_{error}} = \begin{bmatrix} 1 & \delta\beta_3 & -\delta\beta_2 \\ -\delta\beta_3 & 1 & \delta\beta_1 \\ \delta\beta_2 & -\delta\beta_1 & 1 \end{bmatrix} R^{LOS}_{IMU} \qquad \textbf{Equation 15}$$

8 Deprecated Requirements

Requirement(s)	
ST 0801.5-02 (Deprecated)	Floating point values shall comply with IEEE 754.
ST 0801.5-11 (Deprecated)	The definition of slant range shall be as defined in SMPTE RP 210.